

Table 42. EMI characteristics for $f_{HSE} = 16\text{ MHz}$ and $f_{HCLK} = 144\text{ MHz}$

Symbol	Parameter	Conditions	Monitored frequency band	Value	Unit
S_{EMI}	Peak ⁽¹⁾	$V_{DD} = 3.6\text{ V}$, $T_A = 25^\circ\text{C}$, LQFP144 package compliant with IEC 61967-2	0.1 MHz to 30 MHz	15	dB μ V
			30 MHz to 130 MHz	1	
			130 MHz to 1 GHz	27	
			1 GHz to 2 GHz	20	
	Level ⁽²⁾		0.1 MHz to 2 GHz	4.0	-

1. Refer to the EMI radiated test section of the application note EMC design guide for STM8, STM32 and Legacy MCUs (AN1709).
2. Refer to the EMI level classification section of the application note EMC design guide for STM8, STM32 and Legacy MCUs (AN1709).

5.3.12 Electrical sensitivity characteristics

Based on three different tests (ESD, latch-up) using specific measurement methods, the device is stressed in order to determine its performance in terms of electrical sensitivity.

Electrostatic discharge (ESD)

Electrostatic discharges (a positive then a negative pulse separated by 1 second) are applied to the pins of each sample according to each pin combination. The sample size depends on the number of supply pins in the device (3 parts $\times$ (n+1) supply pins). This test conforms to the ANSI/JEDEC standard.

Table 43. ESD absolute maximum ratings

Specified by design and not tested in production.

Symbol	Ratings	Conditions	Packages	Class	Maximum value ⁽¹⁾	Unit
V _{ESD(HBM)}	Electrostatic discharge voltage (human body model)	T _A = 25°C conforming to ANSI/ESDA/JEDEC JS-001	All	2	2000	V
V _{ESD(CDM)}	Electrostatic discharge voltage (charge device model)	T _A = 25°C conforming to ANSI/ESDA/JEDEC JS-002	LQFP 144	C3	750	
			LQFP 100			
			LQFP 80			
			LQFP 64			
			LQFP 48			
			LQFP 32			
			UFQFPN 48			
			UFQFPN 32			

1. Evaluated by characterization - Not tested in production.

Static latch-up

The following complementary static tests are required on three parts to assess the latch-up performance:

- A supply overvoltage is applied to each power supply pin.
- A current injection is applied to each input, output, and configurable I/O pin.

These tests are compliant with EIA/JESD 78E IC latch-up standard.

Table 44. Electrical sensitivities

Symbol	Parameter	Conditions	Class
LU	Static latch-up class	$T_A = 130^\circ\text{C}$ conforming to JESD78	Level II A